

## ● RAG Learning

# Topic 9: Ecosystems and Material Cycles

Edexcel GCSE (9-1) Biology (1BIO) — Paper 2

This topic ties together almost every other area of biology. It asks how living organisms interact with one another and with the non-living world, how energy and matter move through ecosystems, how scientists measure populations in the field, and how human activities are altering the natural cycles that keep the biosphere stable. A strong answer in this topic usually combines three things: (1) precise technical vocabulary, (2) a clear labelled diagram of a cycle or food web, and (3) a worked calculation. The sections below cover each of these in turn.

## 9.1 Key Ecology Terms

Every ecology question in the exam rests on a small set of precise definitions. Learn them word-for-word — examiners routinely award the first mark for an accurate definition of "ecosystem" or "community" and refuse credit for near-misses.

| Term              | Definition                                                                                                        | Example                                              |
|-------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>Ecosystem</b>  | All the living organisms in a given area together with the non-living (abiotic) components and their interactions | A pond, a woodland, a rocky shore                    |
| <b>Community</b>  | All the populations of different species living in the same habitat at the same time                              | All the fish, plants, insects and bacteria in a pond |
| <b>Population</b> | All the individuals of the same species living in the same habitat at the same time                               | All the perch in a lake                              |

|                       |                                                                                              |                                                             |
|-----------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Habitat</b>        | The place where an organism lives                                                            | A rotting log (for a woodlouse); a rock pool (for a limpet) |
| <b>Biotic factor</b>  | A living factor that affects an organism                                                     | Predators, disease, competition, food availability          |
| <b>Abiotic factor</b> | A non-living physical/chemical factor                                                        | Temperature, light, soil pH, water availability             |
| <b>Producer</b>       | An organism that makes its own food by photosynthesis (usually a green plant or alga)        | Grass, oak tree, phytoplankton                              |
| <b>Consumer</b>       | An organism that gets its energy by eating other organisms                                   | Rabbit, fox, human, zooplankton                             |
| <b>Decomposer</b>     | A microorganism (bacterium or fungus) that feeds on dead organic matter, recycling nutrients | Mucor, Pseudomonas, earthworms (detritivore)                |
| <b>Food chain</b>     | A sequence showing who eats what, starting with a producer, with arrows showing energy flow  | Grass → rabbit → fox                                        |
| <b>Food web</b>       | Interlinked food chains showing all the feeding relationships in a community                 | A woodland food web with many chains                        |

**Exam Tip:** Arrows in food chains point **from the organism being eaten to the one eating it**, because they represent the direction of energy/biomass flow — not the direction of the bite.

## 9.2 Abiotic Factors

Abiotic factors are the non-living physical and chemical conditions that an organism experiences. Changes in any one of them can shift a population's size up or down, or move the whole community to a different place.

| Abiotic factor        | Why it matters                                                                                                         | Exam-ready example                                                               |
|-----------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Light intensity       | Drives the rate of photosynthesis in producers, so sets the energy entering the ecosystem                              | Bluebells flower early in spring before the tree canopy closes and light falls   |
| Temperature           | Controls enzyme activity; all metabolic reactions (including photosynthesis and respiration) are temperature-sensitive | A warm spring speeds up insect emergence and plant growth                        |
| Water availability    | Needed for photosynthesis, as a solvent, and for support in plants                                                     | Lack of rain in summer reduces grass growth, so rabbits have less food           |
| Soil pH               | Determines which mineral ions are soluble and which plants can grow                                                    | Heather grows on acidic peat; wild garlic prefers near-neutral woodland soil     |
| Soil mineral content  | Plants need nitrate, phosphate, potassium and magnesium to grow healthy leaves and roots                               | Nitrogen-poor sandy soils support specialist plants (e.g. gorse)                 |
| CO <sub>2</sub> level | Raw material for photosynthesis; limiting at high light intensity                                                      | In a crowded greenhouse, CO <sub>2</sub> concentration may fall and limit growth |
| O <sub>2</sub> level  | Needed for aerobic respiration; critical in aquatic systems where solubility is low                                    | Fish die in a polluted river when dissolved O <sub>2</sub> falls                 |
| Wind intensity        | Increases transpiration; shapes plant growth (e.g. stunted trees on exposed cliffs)                                    | Windward side of a hill has lower plant biomass                                  |

## 9.3 Biotic Factors

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Biotic factors are the effects other living organisms have on a population. The three big ones are competition, predation and disease.

### Competition

Whenever a resource is in short supply, organisms compete for it. Plants compete for **light, water, mineral ions and space**. Animals compete for **food, water, mates, territory and shelter**.

There are two flavours:

- **Intra-specific competition** — between individuals of the *same* species. Typically the strongest form because members of the same species need exactly the same resources. Example: two male red deer competing for mates.
- **Inter-specific competition** — between individuals of *different* species. Example: red squirrels and introduced grey squirrels both competing for acorns in British woodland.

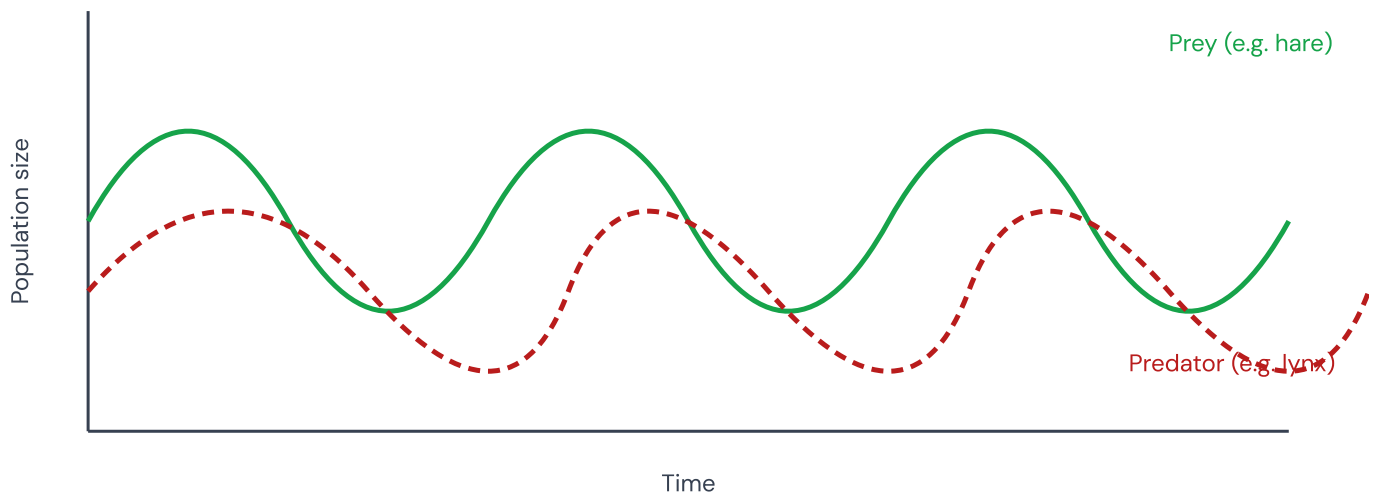
### Predation

Predator and prey populations regulate each other in linked cycles. When prey are abundant, predator numbers rise; this eats down the prey population, which reduces food for predators, so predator numbers fall; prey then recover, and the cycle repeats. Classic example: Canadian lynx and snowshoe hare, with peaks roughly every 10 years.

### Disease

A new pathogen can cause a population crash, especially if individuals are not yet resistant. Example: Dutch elm disease (a fungus) wiped out most British elm trees in the 1970s, also affecting insects that depend on them.

### Predator-prey population cycle



**Exam Tip:** Predator peaks **lag behind** prey peaks — the prey population has to grow first before there is enough food to support more predators.

## 9.4 Interdependence

A community is interdependent: every species relies, directly or indirectly, on many others. Plants rely on bees to pollinate them; foxes rely on rabbits, which rely on grass; fungi rely on dead leaves that were produced by trees that relied on insects eating caterpillars that would otherwise eat the leaves. Because of these links, **removing one species can affect many others**, sometimes triggering a cascade through the web.

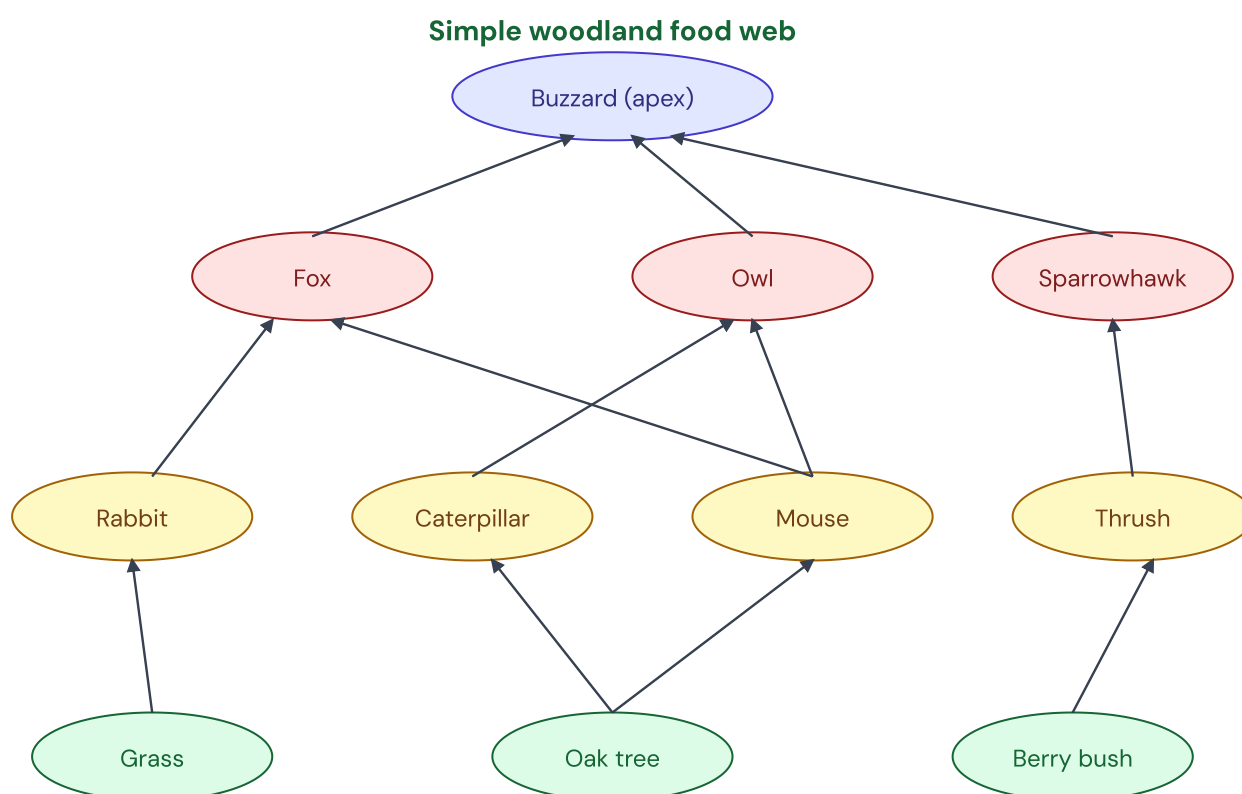
Example: if all the bees in a meadow are killed by a pesticide, the flowers they pollinated produce fewer seeds, the next generation of plants is smaller, the herbivores (rabbits, voles) have less to eat and fall in number, and the carnivores (owls, foxes) lose their food supply. Even the decomposers have less dead material to work with.

## 9.5 Food Chains and Food Webs

A food chain is a simple sequence; real ecosystems are nearly always more complex, with each organism eating (and being eaten by) several others. These interlocking chains form a **food web**.

## Trophic levels

1. **Producer** (level 1) — green plants and algae that use light energy to make organic molecules.
2. **Primary consumer** (level 2) — a herbivore that eats producers (e.g. rabbit).
3. **Secondary consumer** (level 3) — a carnivore that eats primary consumers (e.g. fox).
4. **Tertiary consumer** (level 4) — a carnivore that eats secondary consumers (e.g. eagle).
5. **Quaternary consumer** (level 5) — a carnivore at the top (apex) that eats tertiary consumers.



## 9.6 Pyramids of Numbers, Biomass and Energy

Ecological pyramids show how something decreases at each step up a food chain. Three kinds are used.

| Pyramid | What it shows                               | Behaviour                                                                                            |
|---------|---------------------------------------------|------------------------------------------------------------------------------------------------------|
| Numbers | Number of individuals at each trophic level | Usually a pyramid, but can be <i>inverted</i> (e.g. one oak tree supports thousands of caterpillars) |

|         |                                                                                        |                                                                |
|---------|----------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Biomass | Dry mass of organisms at each trophic level ( $\text{g m}^{-2}$ )                      | Almost always a true pyramid — biomass decreases at each level |
| Energy  | Energy stored in organisms at each trophic level ( $\text{kJ m}^{-2} \text{yr}^{-1}$ ) | Always a true pyramid — energy can only decrease at each level |

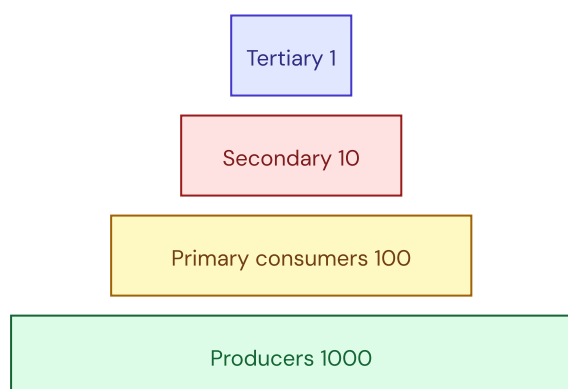
## Why energy decreases at each trophic level

Only about **10% of the energy** available at one trophic level is transferred to the next. The rest is lost:

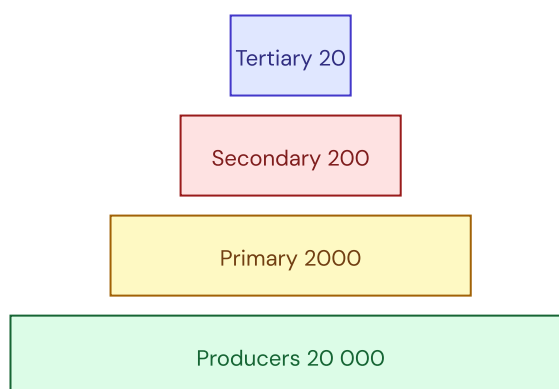
- **Heat from respiration** — most energy from food is released as heat when organisms respire to stay alive, move and grow.
- **Faeces and urine** — some energy leaves in waste that was never absorbed.
- **Uneaten parts** — bones, feathers, roots, bark etc. are usually not eaten and are lost from the chain (to decomposers).

### Pyramid of biomass and pyramid of energy

Biomass ( $\text{g m}^{-2}$ )



Energy ( $\text{kJ m}^{-2} \text{yr}^{-1}$ )



~90% lost at each level: heat (respiration), faeces/urine, uneaten parts

## 9.7 Energy Transfer Efficiency — Worked Example

$$\text{Efficiency (\%)} = (\text{energy transferred to next level} \div \text{energy available at previous level}) \times 100$$

**Worked example:** Grass in a field contains  $20\,000 \text{ kJ m}^{-2} \text{ yr}^{-1}$  of energy. The rabbits that eat the grass contain  $1800 \text{ kJ m}^{-2} \text{ yr}^{-1}$ .

$$\begin{aligned}\text{Efficiency} &= (1800 \div 20\,000) \times 100 \\ &= 9\%\end{aligned}$$

That is close to the 10% rule-of-thumb. Most of the missing 91% has been lost as heat from the rabbits' respiration, in their faeces and urine, or as bits of grass they did not eat.

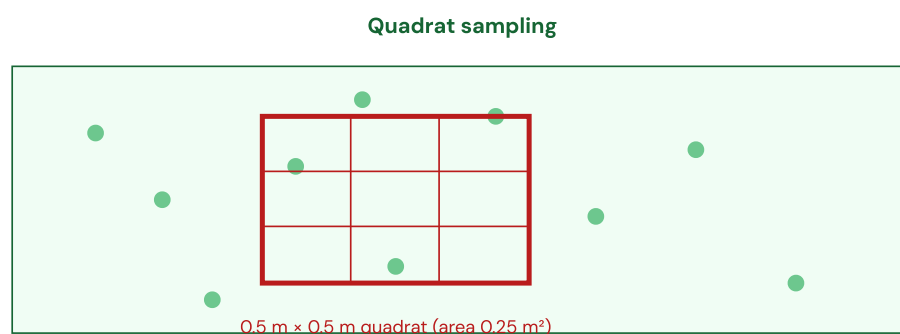
**Exam Tip:** Show the formula *first*, substitute numbers, then state the answer with a **% sign**. Missing the unit loses a mark.

## 9.8 Sampling Techniques

Biologists can't count every organism in a field. Instead, they sample a small representative area and scale up. The method depends on whether the organism moves.

### Quadrats

A **quadrat** is a square frame (typically 0.5 m or 1 m on each side) placed on the ground. Inside it, count the number of individuals of a species (or estimate % cover for plants). Used for plants, slow-moving or sessile animals (e.g. barnacles, limpets).



## Random vs systematic sampling

- **Random sampling** — used when a habitat is fairly uniform. Lay two tape measures at right angles to make a grid; use a random-number generator to pick coordinates; place a quadrat there.
- **Systematic sampling** (transect) — used when conditions change across the habitat. See section 9.10.

## Techniques for mobile animals

| Technique     | What it catches                          | How it works                                                                     |
|---------------|------------------------------------------|----------------------------------------------------------------------------------|
| Pooter        | Small ground insects                     | Suck through a tube; a filter stops insects reaching your mouth                  |
| Sweep net     | Flying insects and insects in long grass | Sweep through vegetation in a figure-of-8                                        |
| Pitfall trap  | Ground-dwelling beetles, spiders         | Buried container with a cover to stop rain; animals fall in and cannot climb out |
| Tree beating  | Insects in tree canopy                   | Hold a sheet beneath a branch, strike with a stick, collect what falls           |
| Kick sampling | Freshwater invertebrates in rivers       | Disturb the river bed with feet while holding a net downstream                   |

## Calculating population size

Estimated population = mean number per m<sup>2</sup> × total area of habitat

**Worked example:** A student places 10 quadrats of 0.25 m<sup>2</sup> randomly in a field of 200 m<sup>2</sup>. Daisy counts per quadrat: 4, 5, 3, 4, 6, 5, 5, 4, 7, 7. Estimate the total number of daisies.

Mean per quadrat =  $(4+5+3+4+6+5+5+4+7+7) \div 10 = 50 \div 10 = 5$   
 Mean per m<sup>2</sup> =  $5 \div 0.25 = 20$  daisies per m<sup>2</sup>

$$\text{Total} = 20 \times 200 = 4000 \text{ daisies}$$

## 9.9 Capture-Recapture (Lincoln Index)

For mobile animals you cannot count directly, use the **Lincoln index**:

$$\text{Estimated population size} = (n_1 \times n_2) \div n_3$$

where  $n_1$  = number caught and marked first time;  $n_2$  = number caught second time;  $n_3$  = number in the second catch that were already marked.

**Worked example:** A biologist catches 60 woodlice and marks them with a tiny dot of non-toxic paint, then releases them. A week later she catches 80 woodlice; 15 are marked.

$$N = (60 \times 80) \div 15 = 4800 \div 15 = 320 \text{ woodlice}$$

**Exam Tip:** Assumptions for the Lincoln index: (1) the mark does not harm the animal or make it easier to catch/see, (2) marked animals mix fully back into the population, (3) no births, deaths or migration between the two catches.

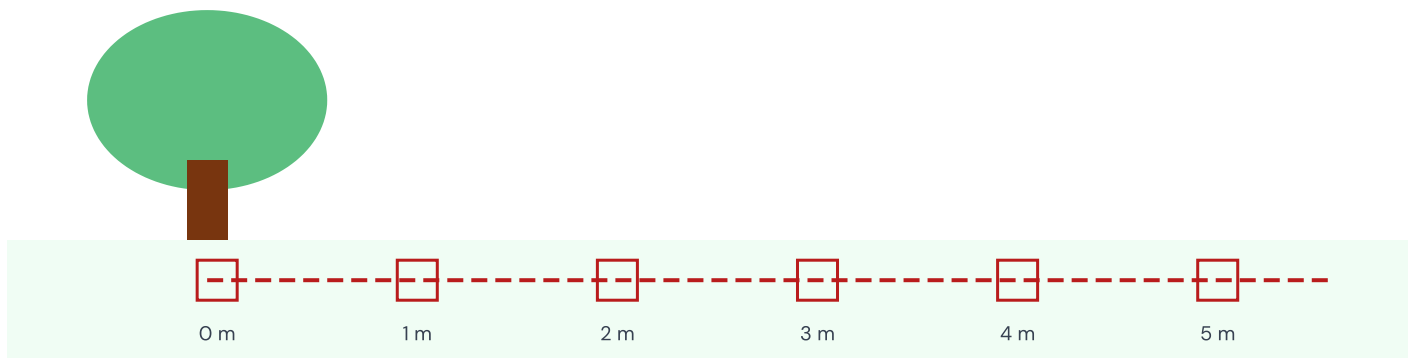
## 9.10 Distribution Along a Transect

When a habitat changes gradually — for example from under a tree canopy to open field — you need to see how distribution changes along the gradient. Lay a **tape measure (transect)** across the gradient and sample at regular intervals.

- **Line transect** — record every species that touches the line at each interval.
- **Belt transect** — place a quadrat at each interval and record % cover or counts.

Example investigation: light intensity vs grass cover along a transect from under a tree out into open field. You would expect grass cover to rise as light intensity rises. Use a **light meter** at each point to measure light intensity (lux) and a quadrat to estimate % grass cover.

### Belt transect from tree canopy into open field



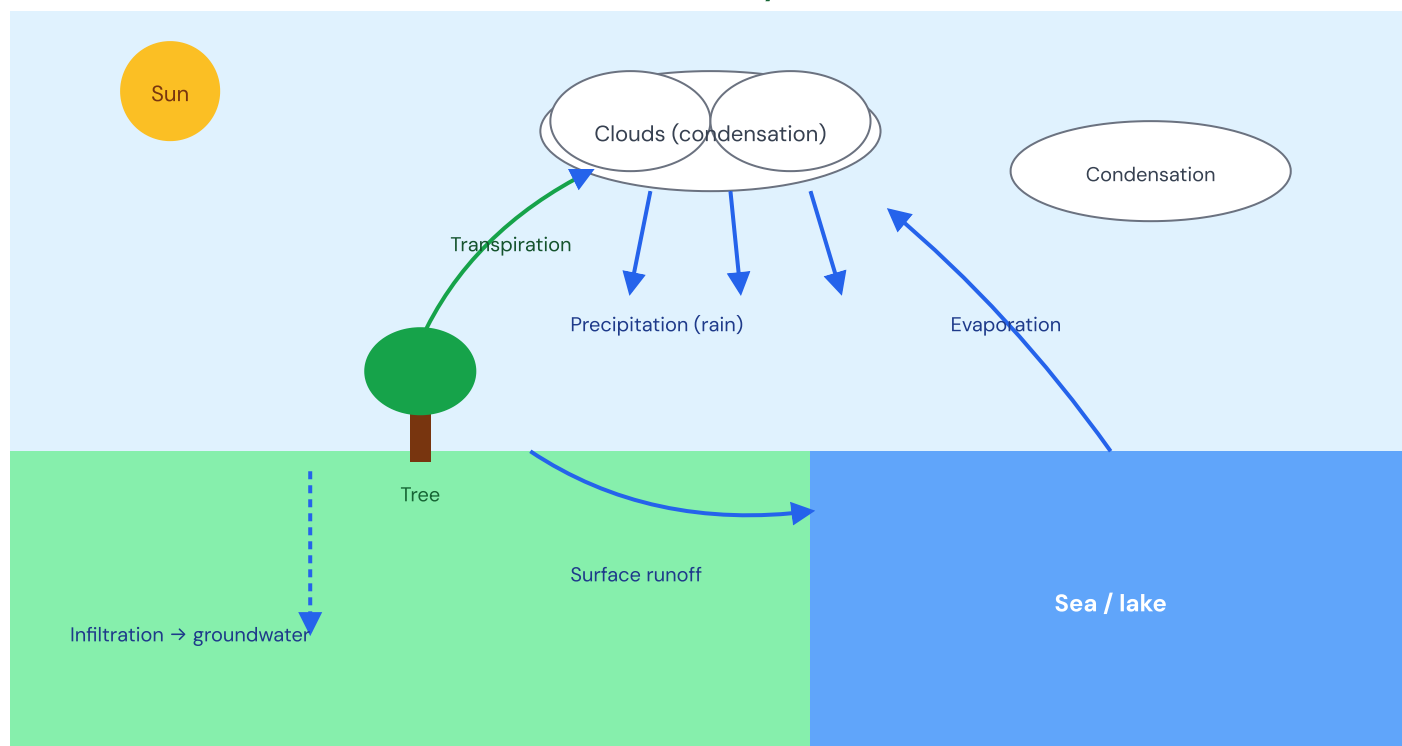
Quadrat placed every metre along the tape; % grass cover and light intensity recorded

## 9.11 The Water Cycle

The water cycle moves water through the ecosystem, making it available for organisms. It is driven by the Sun.

1. **Evaporation** — heat from the Sun turns liquid water in seas, lakes and rivers into water vapour.
2. **Transpiration** — plants lose water vapour from their leaves through the stomata.
3. **Condensation** — as water vapour rises and cools, it condenses into tiny droplets, forming clouds.
4. **Precipitation** — droplets coalesce and fall as rain, snow, sleet or hail.
5. **Surface runoff** — water flows over land into streams, rivers and eventually the sea.
6. **Infiltration / groundwater** — water soaks through soil and rock, forming an underground reservoir that eventually emerges into rivers or seas.

### The water cycle

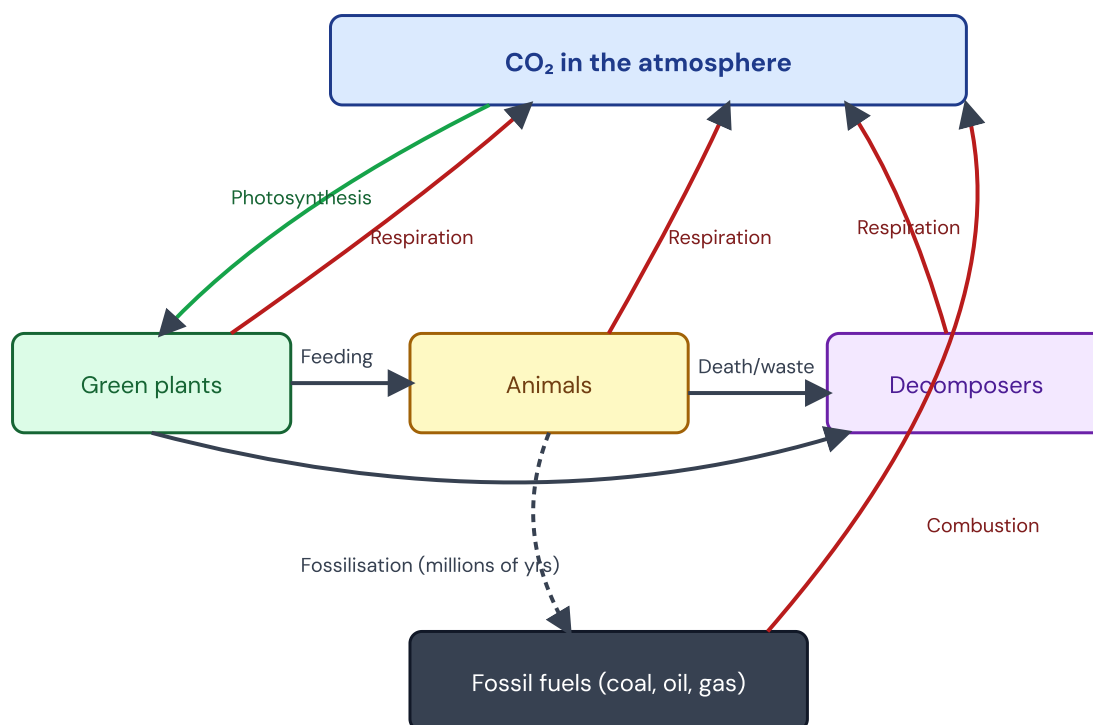


## 9.12 The Carbon Cycle

Carbon is the backbone of every organic molecule. It is continually recycled between the atmosphere, living things, dead material and the oceans.

- **Photosynthesis** removes  $\text{CO}_2$  from the atmosphere: plants (and algae) fix carbon into glucose, which is used to build carbohydrates, proteins and lipids.
- **Feeding** passes carbon along food chains — herbivores eat plants, carnivores eat herbivores.
- **Respiration** in plants, animals and microorganisms releases  $\text{CO}_2$  back to the atmosphere.
- **Decomposition** — decomposers (bacteria, fungi) break down dead organisms and waste; their respiration releases  $\text{CO}_2$ .
- **Combustion** — burning fossil fuels (coal, oil, gas) and wood releases  $\text{CO}_2$  that has been locked away for millions of years.
- **Fossil fuel formation** — dead plants compressed over millions of years under heat and pressure become coal, oil and natural gas.

### The carbon cycle



**Exam Tip:** Every living thing (plant, animal, decomposer) respire — so every living thing releases CO<sub>2</sub>. Only plants and algae remove CO<sub>2</sub> by photosynthesis.

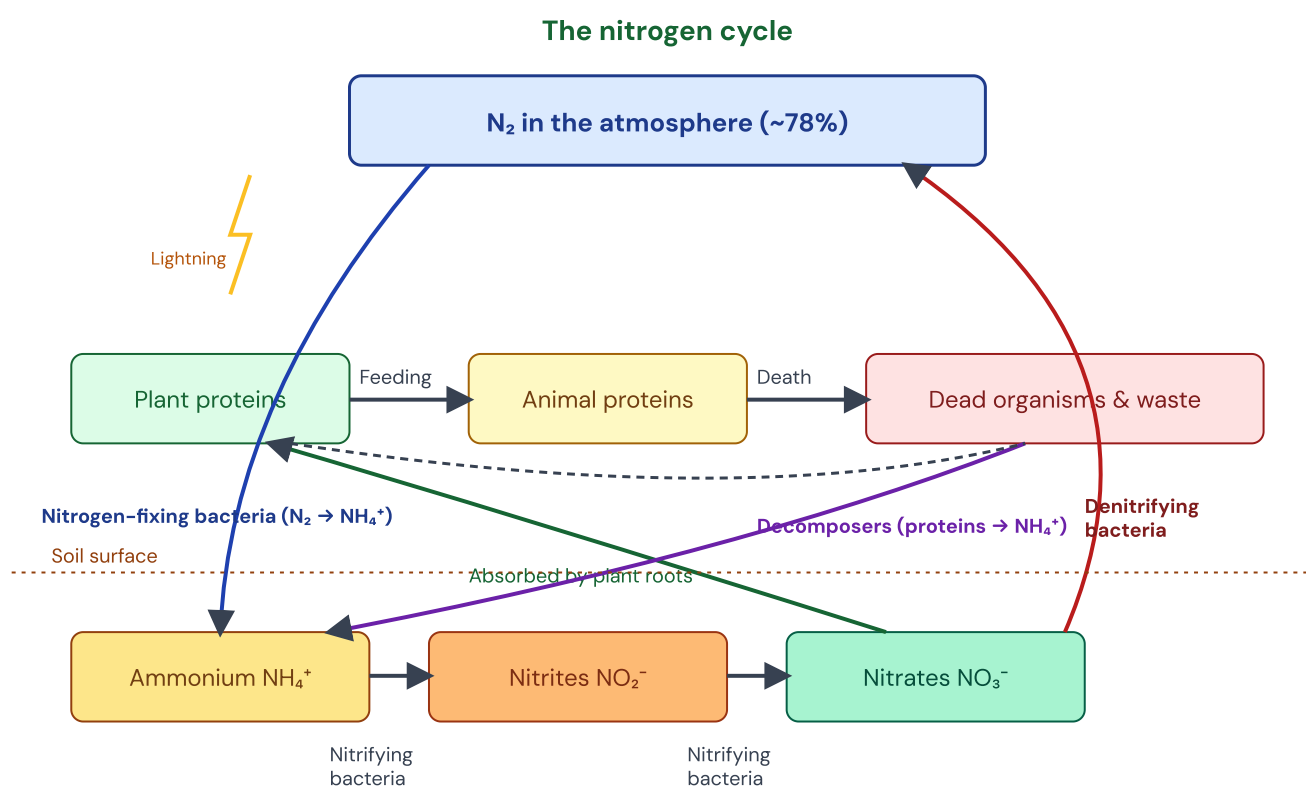
## 9.13 The Nitrogen Cycle

Nitrogen is needed to make proteins and DNA. The atmosphere is ~78% N<sub>2</sub> gas, but N<sub>2</sub> is triple-bonded and unreactive — plants and animals cannot use it directly. **Four types of bacteria** do nearly all the chemical work of converting nitrogen between forms.

| Bacteria                                        | Where                                                        | What they do                                                                                                  |
|-------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Nitrogen-fixing bacteria</b>                 | In soil and in root nodules of legumes (peas, beans, clover) | Convert N <sub>2</sub> gas → ammonium (NH <sub>4</sub> <sup>+</sup> ); these bacteria make nitrogen available |
| <b>Decomposers</b><br>(saprophytic bacteria and | In soil                                                      | Break down proteins in dead organisms and urea in waste →                                                     |

|                              |                                |                                                                                             |
|------------------------------|--------------------------------|---------------------------------------------------------------------------------------------|
| fungi)                       |                                | ammonium                                                                                    |
| <b>Nitrifying bacteria</b>   | In well-aerated soil           | Convert ammonium → nitrites ( $\text{NO}_2^-$ ) → nitrates ( $\text{NO}_3^-$ ) in two steps |
| <b>Denitrifying bacteria</b> | In waterlogged, anaerobic soil | Convert nitrates → $\text{N}_2$ gas — a loss from the soil                                  |

**How plants and animals get their nitrogen:** plant roots absorb nitrates from the soil and use them to make amino acids and proteins. Animals get nitrogen by eating plants (or other animals). Nitrogen returns to the soil in faeces, urine and dead organisms.



**Exam Tip:** Know the four bacteria: nitrogen-fixing, decomposers (saprophytic), nitrifying and denitrifying. A classic 4-mark question asks you to name each and state what it does.

## Why do legumes grow well in poor soil?

Legumes (peas, beans, clover) have a **mutualistic** relationship with nitrogen-fixing bacteria that live inside swellings on their roots called **root nodules**. The bacteria fix atmospheric  $N_2$  into ammonium; the plant supplies the bacteria with sugars from photosynthesis. Farmers often rotate a legume crop through a field to restore soil nitrogen.

## 9.14 Decomposition

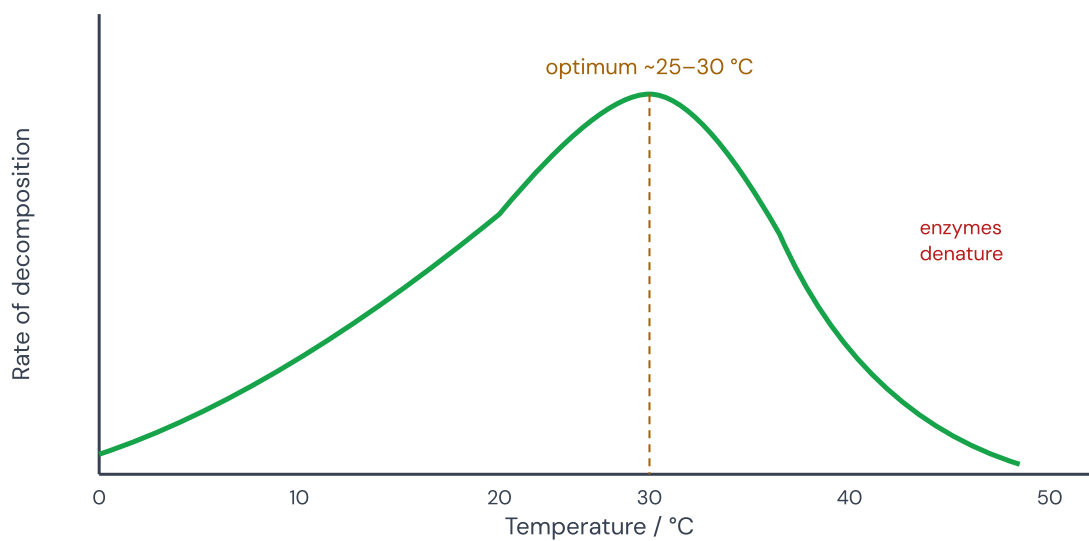
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Decomposers (bacteria, fungi and detritivores such as earthworms, woodlice and maggots) break down dead organic matter and return nutrients to the soil. Without them, the planet would be buried in corpses and nitrogen, phosphorus and carbon would be locked in dead tissue.

### Factors affecting the rate of decomposition

- **Temperature** — decomposers have enzymes with an optimum temperature of about 25–30 °C. Too cold and enzymes work slowly; too hot (above ~45 °C) and enzymes denature. This is why fridges slow food spoilage and why compost heaps warm up.
- **Moisture** — enzymes need water; digestion of dead matter happens outside the decomposer's body (extracellular digestion) in a thin layer of water.
- **Oxygen availability** — most decomposers are aerobic and respire more quickly when  $O_2$  is plentiful; compost heaps are turned to let  $O_2$  in.
- **pH** — most decomposers prefer slightly acidic to neutral soil; very acidic peat bogs preserve bodies for centuries.

### Rate of decomposition vs temperature



## 9.15 Fieldwork on Decomposition

Decomposition rate can be measured in several simple ways in class or field:

- **Fruit slices:** put pieces of apple in sealed bags at different temperatures (fridge, room, warm cupboard) and observe over a week; more fungal growth / softer tissue = faster decomposition.
- **Leaf mass loss:** weigh fresh leaves, put into a mesh bag, leave in soil, dig up weekly and reweigh. Rate = mass lost per day.
- **Compost temperature:** a hot compost heap has lots of respiring decomposers — measuring the temperature rise estimates activity.
- **Milk + lipase experiment:** cresol red indicator turns yellow as lipase breaks down fat to fatty acids — time the colour change at different temperatures.

## 9.16 Human Impacts on Biodiversity

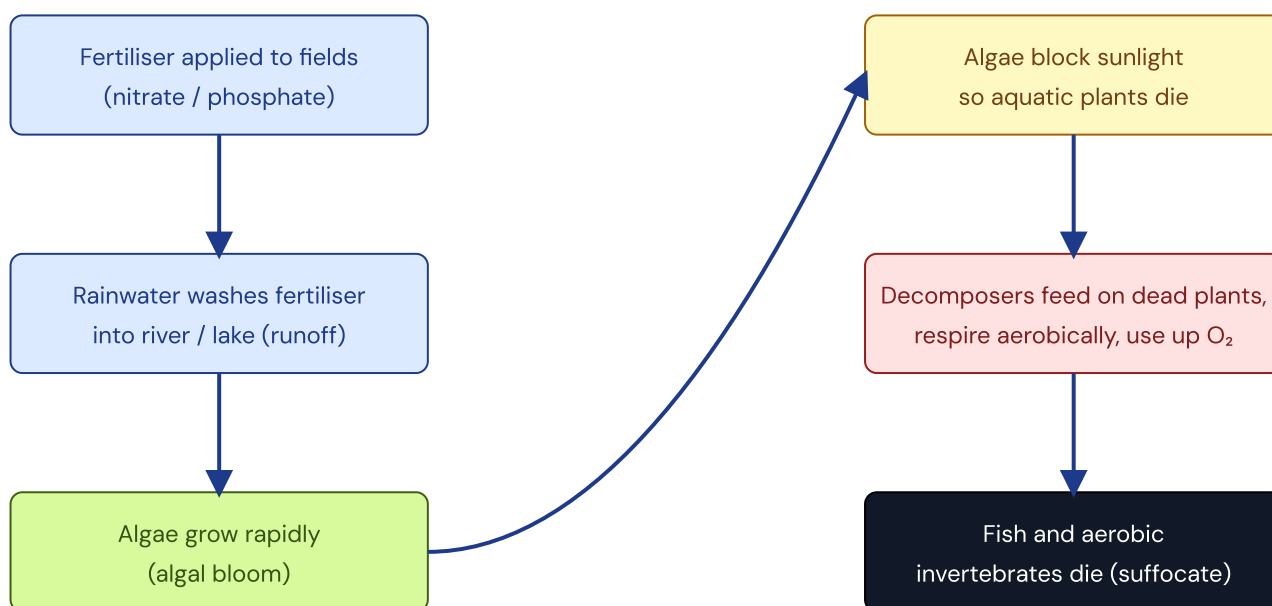
**Biodiversity** is the variety of different species in an ecosystem. High biodiversity makes ecosystems stable: if one species fails, others can fill its niche. Humans reduce biodiversity in many ways:

| Human activity                                                                                                                              | Effect on biodiversity                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Habitat destruction</b> — deforestation for farmland, palm oil, urbanisation                                                             | Loss of habitat → species with nowhere to live → local extinction                        |
| <b>Pollution</b> — air (CO <sub>2</sub> , NO <sub>x</sub> , SO <sub>2</sub> ), water (fertiliser, sewage, oil), land (pesticides, plastics) | Kills sensitive species; causes eutrophication; bioaccumulation of toxins up food chains |
| <b>Overfishing / over-hunting</b>                                                                                                           | Stocks fall below replacement rate → commercial species collapse (e.g. Atlantic cod)     |
| <b>Non-native species</b> — introduced accidentally or deliberately                                                                         | Out-compete natives (grey squirrel vs red squirrel); may carry new diseases              |
| <b>Climate change</b> — driven by greenhouse gas emissions                                                                                  | Species ranges shift; those unable to move or adapt die out (e.g. polar bears)           |

## 9.17 Eutrophication

When fertilisers (nitrates, phosphates) wash off fields into rivers and lakes, a chain reaction damages the freshwater ecosystem.

### Eutrophication pathway

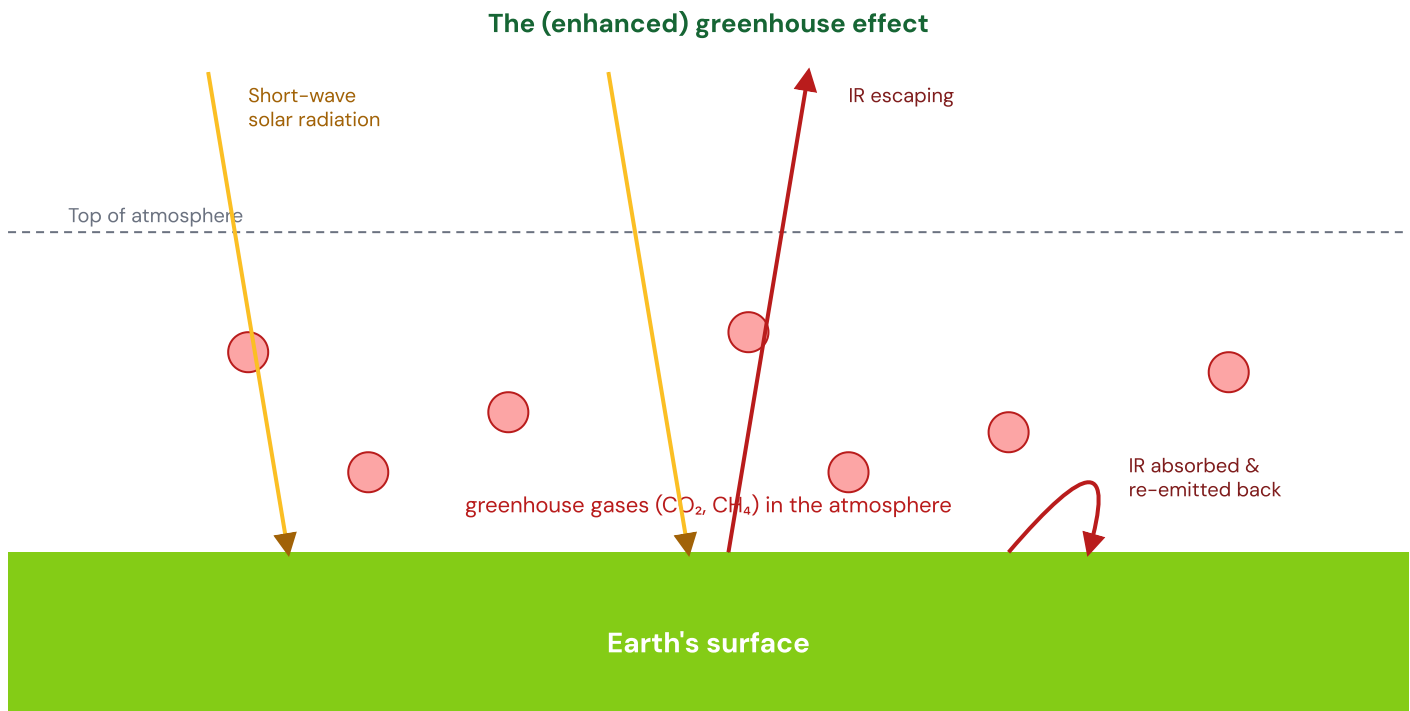


**Exam Tip:** Students often write "algae use up all the oxygen" — wrong. It is the **decomposers (bacteria)** breaking down the dead plant/algal material that use up the O<sub>2</sub> by aerobic respiration.

## 9.18 Greenhouse Gases and Climate Change

Short-wavelength radiation from the Sun passes through the atmosphere and warms the Earth's surface. The Earth re-emits this heat as longer-wavelength **infrared (IR) radiation**. Greenhouse gases (CO<sub>2</sub>, methane CH<sub>4</sub>, water vapour, N<sub>2</sub>O) absorb IR and re-radiate some back towards the surface — this is the **greenhouse effect**.

Human activities (burning fossil fuels, deforestation, intensive livestock farming, paddy fields, landfill) add extra greenhouse gases, **enhancing the natural greenhouse effect** and warming the planet (anthropogenic climate change).



## Consequences of climate change

- Rising sea levels (thermal expansion of water; melting of polar ice caps and glaciers) threaten coastal habitats and cities.
- Shifting species distributions — organisms move towards poles or to higher altitudes as temperatures rise; slow-moving species may not keep up.
- Changes to ecosystems — mismatches in timing (e.g. caterpillars hatching before migrating birds arrive).
- More extreme weather — heatwaves, droughts, floods, storms.
- Ocean acidification from dissolved CO<sub>2</sub> damages coral reefs and shellfish.

## 9.19 Conservation

Conservation is the active protection and management of ecosystems and species. Methods include:

- **Protected habitats** — national parks, nature reserves, Sites of Special Scientific Interest (SSSIs) and marine protected areas where damaging activities are restricted.
- **Captive breeding** — zoos breed endangered species (e.g. Arabian oryx, Californian condor) for later reintroduction.

- **Seed banks** — cold-storing seeds from thousands of plant species (e.g. the Millennium Seed Bank at Kew) as insurance against extinction.
- **Reintroduction programmes** — releasing species back into areas where they were lost (e.g. red kites and beavers in the UK).
- **Reforestation** — planting native trees on cleared land, both to restore habitat and lock up CO<sub>2</sub>.
- **Sustainable fishing quotas** — limiting catch size, setting minimum mesh sizes so young fish can escape, closing nursery grounds.
- **Legislation** — international treaties like CITES banning trade in endangered species and their products.

## Evaluating captive breeding

**Advantages:** keeps genetic stock alive even if wild population collapses; can be reintroduced; raises public awareness.

**Disadvantages:** small gene pool leads to inbreeding; captive-born animals may lack wild survival skills; expensive; reintroduction success rate is variable.

## 9.20 Food Security

**Food security** means all people having reliable access to enough safe, nutritious food. The global population is over 8 billion and still rising, so we must produce more food from the same land — sustainably.

| Technique          | How it increases yield                                                         | Trade-offs                                        |
|--------------------|--------------------------------------------------------------------------------|---------------------------------------------------|
| Selective breeding | Breed cattle with higher milk yields, wheat with more grain per ear            | Reduces genetic diversity; inbreeding problems    |
| Hydroponics        | Grow plants with roots in nutrient solution; precise control; no soil diseases | Expensive set-up; high energy use (pumps, lights) |

|                                       |                                                                                                |                                                                |
|---------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Controlled environments (glasshouses) | Control temperature, CO <sub>2</sub> , light and humidity to optimise photosynthesis           | High fuel/energy costs; high-tech infrastructure               |
| Biological pest control               | Use a natural predator instead of pesticide (e.g. ladybirds vs aphids)                         | Predator itself may become a pest if non-native; slower to act |
| Fertilisers                           | Replace nitrates, phosphates and potassium removed by crops                                    | Runoff causes eutrophication; fossil fuels used to make them   |
| Pesticides / herbicides               | Kill pests and weeds, reducing competition and damage to crops                                 | Harm non-target species; bioaccumulation; resistance evolves   |
| Genetic modification                  | Insert genes for pest-resistance, drought-tolerance, higher vitamin content (e.g. golden rice) | Expensive; public concern; possible spread to wild populations |
| Limiting human food chain length      | Fewer energy losses if humans eat plants directly rather than feeding crops to animals first   | Cultural and nutritional constraints on vegetarian diets       |
| Intensive farming of livestock        | Restricting movement & keeping warm reduces energy loss as heat, increases mass gain           | Welfare concerns; higher disease risk; antibiotic use          |

## 9.21 Why Biodiversity Matters

Maintaining biodiversity is not just an ethical preference — it keeps ecosystems (and humans) functioning.

- **Ecosystem stability** — diverse communities are more resilient to disease, pests, and climatic variation.

- **Ecosystem services** — pollination by insects, water purification by wetlands, flood defence by mangroves, nutrient recycling by decomposers.
- **Medicinal discovery** — many drugs come from plants or microbes (aspirin from willow bark, penicillin from *Penicillium*, quinine from cinchona tree). Losing species is losing potential medicines.
- **Food security** — wild relatives of crops carry genes (for disease resistance, drought tolerance) that future plant breeding depends on.
- **Cultural and aesthetic value** — wild places support tourism, recreation and wellbeing.
- **Moral responsibility** — many people believe we have a duty to other species.

**Exam Tip:** For long-answer questions ("Explain why biodiversity should be maintained"), give **at least three distinct reasons** and support each with a concrete example — this is how top-band answers earn full marks.

## End-of-topic summary

| Spec      | Key idea                                                                                                                                   |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 9.1–9.4   | Ecosystems are communities of populations interacting with abiotic factors; biotic and abiotic factors together determine population size  |
| 9.5–9.7   | Food chains/webs show energy flow; biomass and energy fall ~90% at each step, mainly because of respiration heat, faeces and uneaten parts |
| 9.8–9.10  | Quadrats and transects for plants; capture-recapture (Lincoln index) for mobile animals; population = mean per m <sup>2</sup> × area       |
| 9.11–9.13 | Water, carbon and nitrogen are recycled; four kinds of bacteria drive the nitrogen cycle                                                   |
| 9.14–9.15 | Decomposition is enzyme-based — affected by temperature, moisture, O <sub>2</sub> and pH                                                   |

|               |                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------|
| 9.16–<br>9.18 | Human impact reduces biodiversity; eutrophication and climate change are major threats        |
| 9.19–<br>9.21 | Conservation, sustainable food production and biodiversity all matter for long-term stability |